GazeMamba: Inter-Gaze State-Space Modeling for Strabismus Subtype Classification

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Abstract. Classification of strabismus subtypes from the photograph of the nine cardinal positions of gaze is commonly formulated as static image recognition, which treats nine-gaze photographs as independent spatial observations and overlooks the dependencies induced by eye movement during acquisition. This limitation is critical because subtype-discriminative evidence is often reflected not only in individual gaze images but also in directional changes of ocular alignment across gaze positions. Existing sequence modeling strategies, such as raster-order scanning and unordered aggregation, fail to preserve the intrinsic acquisition order of the nine-gaze examination by introducing artificial spatial adjacency or discarding directional relations among gaze states. To address this issue, we propose **GazeMamba**, an acquisition-ordered framework that explicitly models gaze-to-gaze dependencies through state-space sequence modeling. A shared gaze-wise encoder first converts nine-gaze photographs into semantic tokens, which are then processed by two residual Mamba blocks following the authentic acquisition sequence and decoded through dual diagnostic heads. By separating gaze-specific appearance from inter-gaze state transitions, GazeMamba captures clinically relevant cross-gaze variation without imposing arbitrary spatial assumptions. We establish a six-subtype benchmark comprising 1,716 nine-gaze photographs and evaluate static, pooled, order-aware, and structural alternatives under a fixed held-out protocol. GazeMamba consistently outperforms the strongest same-protocol composed-image baseline, improving accuracy and weighted F1 by 1.17 and 3.20 percentage points, respectively. Ablation studies further demonstrate that acquisition-ordered linear sequence modeling provides more reliable subtype discrimination than unordered pooling or more complex geometric modeling strategies.

Keywords: Strabismus Subtype classification · State-space model · Mamba · Medical Image Processing

1 Introduction

Strabismus is an ocular-misalignment disorder that can impair binocular vision and visual development when it is not identified and treated appropriately. Automated analysis has therefore been investigated using facial photographs, ocular landmarks, corneal-reflection geometry, and deep neural networks [2,7,9,10,21]. The photograph of the nine cardinal positions of gaze provides richer evidence than a primary-gaze image because each position reveals ocular alignment under a different extraocular-muscle configuration.

This task is fundamentally relational. A subtype cannot always be characterized by the appearance of one gaze crop alone. In A- and V-pattern strabismus, for example, the diagnostic cue is the change in horizontal deviation between elevation and depression. A composed-image CNN or Transformer [1,5,19] may absorb such relations implicitly, but its representation does not distinguish within-gaze appearance from between-gaze change. Conversely, permutation-invariant pooling and set-based aggregation [11,17,22] deliberately discard order. The resulting representation can be insensitive to directional changes between gaze states and may therefore underrepresent pattern-related evidence.

Recent structured methods have already demonstrated that cross-gaze relations are valuable. CI-GNN organizes clinically derived ocular variables as a graph, whereas DECIS combines intra-gaze and inter-gaze evidence with diagnostic rules [18,23]. Our contribution is complementary rather than a claim that previous work ignores inter-gaze structure. We directly model gaze-wise visual tokens with a state-space sequence encoder and investigate how the semantic ordering of those tokens affects classification.

To this end, we propose **GazeMamba**. The photograph is decomposed into nine gaze-specific crops, which are encoded independently by a shared visual backbone. As illustrated in **Figure 1**, image acquisition begins at the primary gaze position (position 5), after which the patient is instructed to move the eyes through the remaining gaze positions following the trajectory

$$5 \to 2 \to 3 \to 6 \to 9 \to 8 \to 7 \to 4 \to 1.$$

The resulting gaze-level representations are arranged according to this trajectory and processed by residual Mamba blocks.

From a clinical perspective, the nine-gaze examination is used to assess incomitance, namely, how ocular deviation varies across gaze directions under different extraocular-muscle configurations. The acquisition protocol begins at the primary gaze position and then traverses the peripheral gaze positions along a continuous path. Consequently, consecutive positions are generally anatomically neighboring gaze states, and each transition changes only one horizontal or vertical component of the gaze direction. This organization enables the sequence encoder to compare ocular alignment under progressively changing extraocular-muscle configurations. Such comparisons are particularly relevant to A- and V-pattern strabismus because their diagnostic evidence is defined by directional variation in deviation rather than by the appearance of an individual gaze view.

Accordingly, the acquisition trajectory provides a clinically motivated semantic ordering rather than merely a temporal record of image capture. It preserves local transitions between neighboring gaze states, whereas row-major traversal may introduce discontinuous transitions between the end of one row and the beginning of the next, and permutation-invariant aggregation removes directional relations altogether. We therefore hypothesize that propagating information along the acquisition trajectory provides a more coherent representation of inter-gaze variation. This ordering should not be interpreted as the diagnostic reading order of an ophthalmologist or as a universally optimal clinical traversal. Instead, it is a protocol-specific inductive bias whose effectiveness must be established empirically. We therefore compare it with row-major, reverse-acquisition, fixed-random, closed-loop, circular, bidirectional, and center-ring alternatives.

Unlike visual Mamba backbones that apply state-space modeling to serialized image patches, GazeMamba operates at the gaze level, with each token representing one complete gaze view [3, 14, 24]. This design separates gaze-specific visual appearance from inter-gaze state transitions and allows information to be selectively propagated across semantically ordered gaze positions.

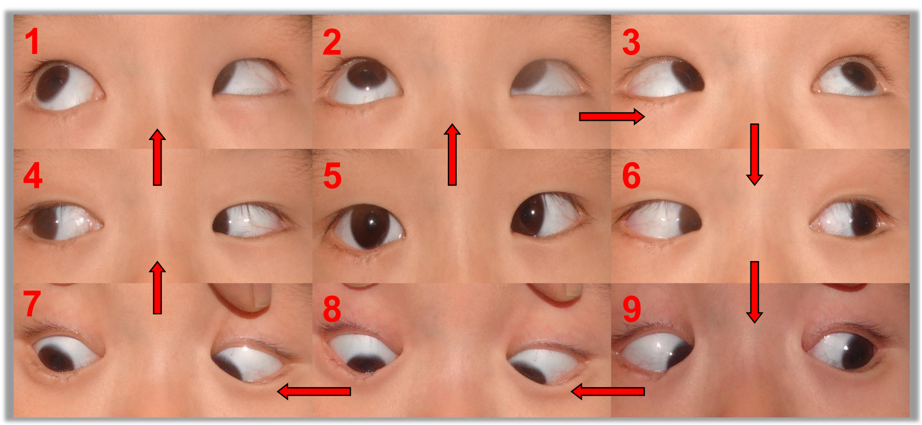

Figure 1. Acquisition trajectory for the photograph of the nine cardinal positions of gaze. Image acquisition begins at the primary gaze position (position 5) and follows the sequence $5 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 9 \rightarrow 8 \rightarrow 7 \rightarrow 4 \rightarrow 1$. The numbered regions denote the nine gaze positions, and the red arrows indicate transitions between consecutively acquired ocular states. This ordering represents the patient's eye-movement trajectory during image acquisition rather than the diagnostic reading order of an ophthalmologist.

In collaboration with ophthalmologists at the Joint Shantou International Eye Center of Shantou University and the Chinese University of Hong Kong, we curate a clinically annotated evaluation benchmark comprising 1,716 photographs of the nine cardinal positions of gaze. Each photograph is assigned a fine-grained strabismus subtype label by senior ophthalmologists. The benchmark covers six clinically defined categories combining deviation type with A/V-pattern characteristics, including dissociated vertical deviation, esotropia, and exotropia with or without the corresponding pattern characteristics. We establish a fixed class-stratified training, validation, and held-out test protocol to support controlled evaluation of composed-image classifiers, unordered multi-gaze aggregation, order-aware sequence models, and geometry-aware sequence constructions. Detailed label definitions, cohort information, ethical approval, and experimental settings are provided in the Experiments section.

On the present dataset, GazeMamba improves accuracy by 5.26 percentage points and weighted F1 by 9.21 percentage points over unordered mean pooling when both methods use identical gaze-wise features. The complete model achieves 67.25% accuracy and 63.88% weighted F1. Acquisition ordering gives the strongest single-run result, although its multi-seed margin over row-major ordering is modest. We therefore interpret the evidence as support for explicit inter-gaze modeling rather than proof of a universally optimal traversal.

Our contributions are threefold:

- We formulate classification from the photograph of the nine cardinal positions of gaze as gaze-level semantic sequence modeling, explicitly separating within-gaze appearance from inter-gaze variation.
- We introduce GazeMamba, a backbone-agnostic framework that applies selective state-space modeling to gaze-level tokens arranged along a clinically motivated acquisition trajectory.
- We establish a clinically annotated benchmark of 1,716 nine-gaze photographs covering six strabismus subtypes and systematically evaluate unordered aggregation, alternative gaze orders, and structural sequence variants.

2 Related Work

2.1 Automated Strabismus Analysis

Automated strabismus analysis has progressed from corneal-reflection and landmark geometry to learning-based screening and classification. Existing systems analyze primary-gaze facial photographs, estimate ocular landmarks, or quantify deviation from photographs of multiple gaze positions [2, 7, 9, 10, 21]. These studies establish that visual evidence can support automated assessment, but most image-level models do not explicitly separate view-specific appearance from changes across gaze states.

2.2 Multi-View and Set Representation Learning

Multi-view learning commonly combines observations by averaging, feature concatenation, recurrent modeling, or attention [8]. Deep Sets and Set Transformer provide principled permutation-invariant representations [11, 22], while multi-set Transformers extend attention to interactions among several sets [17]. These models are appropriate when view order is irrelevant. The nine-gaze examination differs because each position has a fixed anatomical meaning and directional changes between positions may contain subtype evidence. We therefore use mean pooling as an order-free control and compare it with explicit sequence modeling.

2.3 Structured and Interpretable Strabismus Modeling

Recent studies have introduced structured clinical representations into strabismus classification. Zheng et al. [23] proposed CI-GNN, which selects causally relevant ocular variables and organizes the nine gaze positions as a graph for interpretable subtype prediction. Tang et al. [18] developed DECIS to combine intra-gaze and inter-gaze visual evidence with clinical rules for verifiable diagnostic reasoning. These methods confirm the importance of cross-gaze structure. GazeMamba addresses a different question by operating directly on gaze-wise visual tokens and testing whether state propagation along a semantic gaze sequence improves subtype recognition.

2.4 State-Space Models for Vision

Structured state-space models provide efficient long-range sequence modeling [3, 4]. Vision Mamba and VMamba adapt this family to image representation using bidirectional or multi-directional patch scanning [14, 24]. Related work has explored deformable scanning, efficient hidden-state mixing, and biomedical segmentation [12, 13, 16, 20]. GazeMamba does not introduce another patch-scanning backbone. It operates at a higher semantic level in which each token represents one complete gaze view and the scan order represents a hypothesized relation among gaze states.

3 Method

3.1 Problem Formulation

Let G denote the number of standardized gaze positions and let $I_g \in \mathbb{R}^{C\times H\times W}$ be the crop observed at position $g \in \{1, \ldots, G\}$. A sample is represented by $\mathcal{X} = (I_1, \ldots, I_G)$ with class label $y \in \mathcal{Y}$. The objective is to learn a classifier F that uses both gaze-specific appearance and relations among gaze states:

$$\hat{y} = F(\mathcal{X}, \pi),$$

where π is a permutation that specifies the semantic order in which gaze positions are presented to the sequence encoder. The main model uses the acquisition trajectory $\pi_a = (5, 2, 3, 6, 9, 8, 7, 4, 1)$. Alternative permutations are evaluated as controlled order variants rather than assumed to be clinically optimal.

3.2 Gaze-Wise Visual Encoding

The same visual encoder E_θ is applied to every gaze crop, producing

$$\mathbf{v}_g = E_\theta(I_g) \in \mathbb{R}^{d_v},$$

where θ denotes the shared encoder parameters and d_v is the gaze-feature dimension. Weight sharing ensures that differences among $\mathbf{v}_g$ arise from gaze content rather than position-specific encoders.

For a permutation π, the feature at sequence step t is projected into the state-space dimension d:

$$\mathbf{z}_t = W_e \mathbf{v}_{\pi(t)} + \mathbf{b}_e + \mathbf{p}_t,$$

where $W_e \in \mathbb{R}^{d \times d_v}$ and $\mathbf{b}_e \in \mathbb{R}^d$ are learnable projection parameters, and $\mathbf{p}_t \in \mathbb{R}^d$ is a learnable sequence-position embedding. The ordered token matrix is $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_G]^\top \in \mathbb{R}^{G \times d}$.

3.3 Inter-Gaze State-Space Modeling

The ordered token matrix is processed by L residual Mamba blocks. Let $\mathbf{H}^{(0)} = \mathbf{Z}$. The output of block $\ell \in \{1, \dots, L\}$ is

$$\mathbf{H}^{(\ell)} = \mathbf{H}^{(\ell-1)} + \mathcal{M}_\ell\Big(\mathrm{LN}_\ell(\mathbf{H}^{(\ell-1)})\Big),$$

where LN_ℓ denotes layer normalization and $\mathcal{M}_\ell$ denotes the selective state-space transformation. To make the state propagation explicit, let $\mathbf{x}_t$ be the normalized token at step t. A selective state-space layer updates its latent state and produces an output according to

$$\boldsymbol{\delta}_t, \mathbf{B}_t, \mathbf{C}_t = f_\ell(\mathbf{x}_t), \qquad \bar{\mathbf{A}}_t = \exp(\boldsymbol{\delta}_t \mathbf{A}),$$
$$\mathbf{s}_t = \bar{\mathbf{A}}_t \mathbf{s}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t, \qquad \mathbf{m}_t = \mathbf{C}_t \mathbf{s}_t + \mathbf{D}\mathbf{x}_t,$$

where the step size $\boldsymbol{\delta}_t$ and the input/output projections $\mathbf{B}_t$ and $\mathbf{C}_t$ depend on the current token, and $\bar{\mathbf{B}}_t$ is the corresponding discretized input matrix. Consequently, the model can selectively retain or suppress information propagated from preceding gaze states. This mechanism explicitly represents gaze-to-gaze variation without concatenating all views into one spatial image.

After the final block, normalization and sequence mean pooling produce

$$\mathbf{h} = \frac{1}{G} \sum_{t=1}^{G} \mathrm{LN}_{\mathrm{out}}\Big(\mathbf{H}_t^{(L)}\Big).$$

Here $\mathrm{LN}_{\mathrm{out}}$ is the layer normalization applied after the final residual block, $\mathbf{H}_t^{(L)} \in \mathbb{R}^d$ is the final hidden representation at step t, and $\mathbf{h} \in \mathbb{R}^d$ is the sample-level representation. Unlike visual Mamba methods that serialize image patches, GazeMamba defines each token as a complete gaze view. The state transition therefore operates between clinically interpretable gaze states.

3.4 Dual-Head Diagnostic Decoding

The diagnostic decoder factorizes the label into deviation type and pattern attributes. Given $\mathbf{h}$, the two heads produce

$$\mathbf{o}^{\text{type}} = W_t\mathbf{h} + \mathbf{b}_t, \qquad \mathbf{o}^{\text{pattern}} = W_r\mathbf{h} + \mathbf{b}_r,$$

where W_t and W_r are classifier matrices, $\mathbf{b}_t$ and $\mathbf{b}_r$ are biases, and the two logit vectors predict deviation type and pattern, respectively. For ground-truth attributes y^{type} and y^{pattern}, the training objective is

$$\mathcal{L} = \mathrm{CE}(\mathbf{o}^{\text{type}}, y^{\text{type}}) + \mathrm{CE}(\mathbf{o}^{\text{pattern}}, y^{\text{pattern}}).$$

Here CE denotes the cross-entropy loss. At inference, let $\mathcal{C}$ be the set of six valid type–pattern pairs. For a legal class $c = (c_t, c_r) \in \mathcal{C}$, its joint score is

$$s(c) = \log\mathrm{softmax}(\mathbf{o}^{\text{type}})_{c_t} + \log\mathrm{softmax}(\mathbf{o}^{\text{pattern}})_{c_r}, \qquad \hat{y} = \arg\max_{c\in\mathcal{C}} s(c).$$

Thus, the two heads contribute additive log-probabilities, and the highest-scoring legal combination is returned as the six-class prediction.

Figure 2 summarizes the complete processing path from gaze-wise encoding to state-space modeling and factorized prediction. The design is independent of a particular visual backbone or feature dimension. Concrete implementation choices are reported in Section 4.1.

4 Experiments

4.1 Dataset, Protocol, and Implementation Details

Dataset and ethics. The dataset contains 1,716 photographs of the nine cardinal positions of gaze collected at the Joint Shantou International Eye Center of Shantou University and the Chinese University of Hong Kong. Each photograph was annotated by senior ophthalmologists with a fine-grained strabismus subtype label. The labels comprise six categories: dissociated vertical deviation without A/V pattern, esotropia without A/V pattern, esotropia with V pattern, exotropia with A pattern, exotropia without A/V pattern, and exotropia with V pattern.

This retrospective study was approved by the Ethics Committee of the Joint Shantou International Eye Center of Shantou University and the Chinese University of Hong Kong (approval no. EC 20260325(2)-P06) and was registered with the Chinese Clinical Trial Registry (ChiCTR; registration no. ChiCTR2600122191). A waiver of informed consent was granted because the study used a retrospective cohort of de-identified images. A class-stratified 70/20/10 split produced 1,202 training samples, 343 validation samples, and 171 held-out test samples. Checkpoint selection was performed exclusively on the validation set.

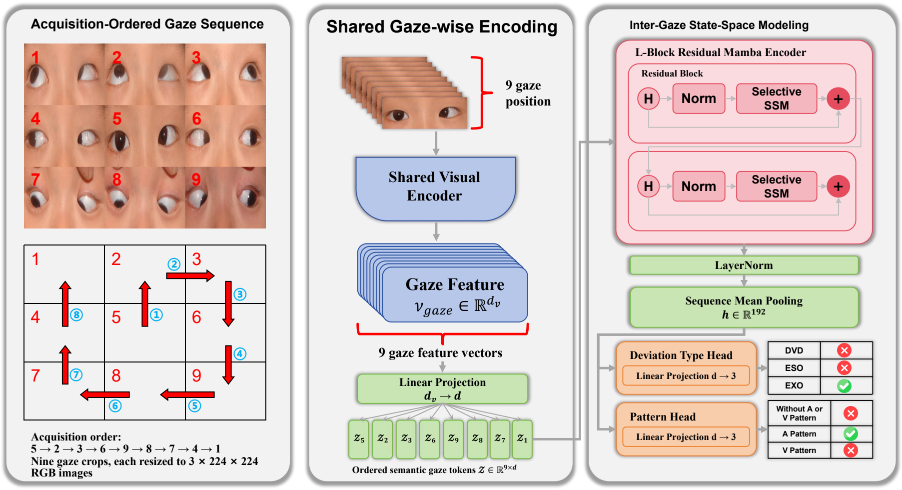

Figure 2. Overall framework of GazeMamba. A shared visual encoder converts gaze-specific crops into semantic tokens. The tokens are arranged according to a gaze sequence and processed by residual Mamba blocks. Sequence pooling produces the common representation used by the deviation-type and pattern heads.

Implementation details. Each gaze crop is resized to 224×224. The main implementation uses an ImageNet-pretrained DenseNet-121 as the shared visual encoder, producing $d_v = 1024$ features. The projection and state-space dimensions are $d = 192$ and $d_{\text{state}} = 16$, respectively. We use $L = 2$ residual Mamba blocks with dropout 0.1. AdamW uses a learning rate of 2×10^{-4} and weight decay of 0.05. During end-to-end training, the visual backbone is fine-tuned at 2×10^{-5} and the remaining modules at 2×10^{-4} with cosine annealing for 100 epochs. Composed-image baselines receive the complete photograph, whereas GazeMamba receives nine gaze-specific crops. Order-sensitive comparisons are repeated with three random initializations. The fixed random ordering is generated once and kept unchanged across otherwise identical runs. The complete-model comparison uses end-to-end fine-tuning, whereas the controlled sequence-modeling, gaze-ordering, and structural ablations use identical frozen gaze-wise features across all compared methods. Their absolute performance values should therefore not be directly compared. The frozen-feature experiments are designed specifically to isolate the effects of inter-gaze modeling and sequence construction.

4.2 Comparison with Visual Baselines

In the full-data experiment, all visual baselines and GazeMamba are trained using the complete 1,202-sample training split, with 343 samples for validation and 171 samples for held-out testing. ViT-Base, Swin-B, and DenseNet-121 perform end-to-end classification directly from the composed photograph of the nine cardinal positions of gaze. GazeMamba instead divides the photograph into nine gaze-wise crops, encodes them with a shared DenseNet-121, and models their acquisition order using two residual Mamba blocks. We additionally re-evaluate CI-GNN [23] using its official feature definitions, graph architecture, and dual classification heads. Its deviation-type and A/V-pattern outputs are combined into a six-subtype prediction that is correct only when both heads are correct.

Table 1. Comparison of visual baselines, CI-GNN, and GazeMamba.

Method	Accuracy (%)	Weighted F1 (%)	Macro F1 (%)
ViT-Base [1]	59.65	53.34	28.78
Swin-B [15]	63.16	52.72	31.39
DenseNet-121 [6]	66.08	60.68	43.38
CI-GNN [23]	56.73	51.17	28.95
GazeMamba (Ours)	**67.25**	**63.88**	**47.22**

As shown in **Table 1**, DenseNet-121 achieves the strongest performance among conventional visual baselines, obtaining 66.08% accuracy, 60.68% weighted F1, and 43.38% macro F1. GazeMamba further improves these results by 1.17, 3.20, and 3.84 percentage points, respectively, demonstrating that gaze-wise encoding and acquisition-ordered state-space modeling can capture additional discriminative information beyond static composed-image classification. By explicitly preserving the sequential dependencies among gaze positions, GazeMamba better models the inter-gaze variations associated with strabismus subtypes and achieves more balanced recognition across categories.

The CI-GNN baseline achieves relatively limited performance under the same held-out evaluation protocol. This suggests that directly modeling spatial relationships among gaze positions may not be sufficient to capture the directional transition patterns contained in the acquisition sequence. In contrast, GazeMamba leverages the inherent ordering of gaze acquisition and sequential state transitions, leading to improved feature representation and more robust subtype discrimination.

4.3 Effectiveness of Inter-Gaze Modeling

To isolate sequence modeling from visual encoding, we compare GazeMamba with mean pooling using identical cached gaze-wise features. Mean pooling treats the nine representations as an unordered set, whereas GazeMamba propagates information across the ordered gaze states. The visual features and classifier supervision remain unchanged.

As shown in **Table 2**, GazeMamba improves accuracy by 5.26 percentage points, weighted F1 by 9.21 points, and macro F1 by 9.90 points. Because the

Table 2. Controlled comparison of unordered aggregation and inter-gaze state-space modeling using identical frozen gaze-wise features.

Method	Accuracy (%)	Weighted F1 (%)	Macro F1 (%)
Mean pooling	54.39	45.55	19.98
Ours	**59.65**	**54.76**	**29.88**

gaze-wise visual features are unchanged, these gains provide empirical evidence for the benefit of order-sensitive inter-gaze modeling under fixed gaze-wise visual features.

4.4 Effect of Gaze Ordering

Building on the explicit sequence-modeling gain established in **Table 2**, we next test whether sequence definition itself matters. Four orders are evaluated in **Figure 3**: acquisition, row-major $(1, 2, \ldots, 9)$, reverse acquisition, and a fixed random permutation. All other model and training settings are unchanged.

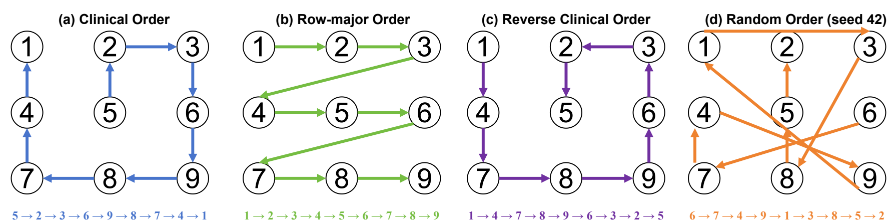

Figure 3. Semantic orderings considered for the same nine gaze positions. Acquisition order follows the patient's standardized eye-movement trajectory, whereas row-major, reverse, and fixed random orders provide controlled alternatives.

The acquisition trajectory gives the strongest representative result, whereas reversing that trajectory is least effective. This direction sensitivity is consistent with the selective state update, which propagates information according to token order. The acquisition path also follows the observed succession of ocular states and avoids the artificial row breaks introduced by raster traversal. To assess whether this observation is robust to random initialization, we additionally repeat the acquisition-order and row-major experiments using three random seeds. Nevertheless, across these three initializations, the mean accuracy of acquisition ordering exceeds that of row-major ordering by only 0.59 percentage points. The evidence therefore supports order-aware inter-gaze modeling more strongly than a claim that one traversal is universally optimal.

Table 3. Order ablation using the same GazeMamba architecture and gaze-wise features.

Ordering	Accuracy (%)	Weighted F1 (%)
Acquisition (Ours)	**59.65**	**54.76**
Row-major	56.14	53.97
Reverse acquisition	53.80	48.66
Fixed random	58.48	52.37

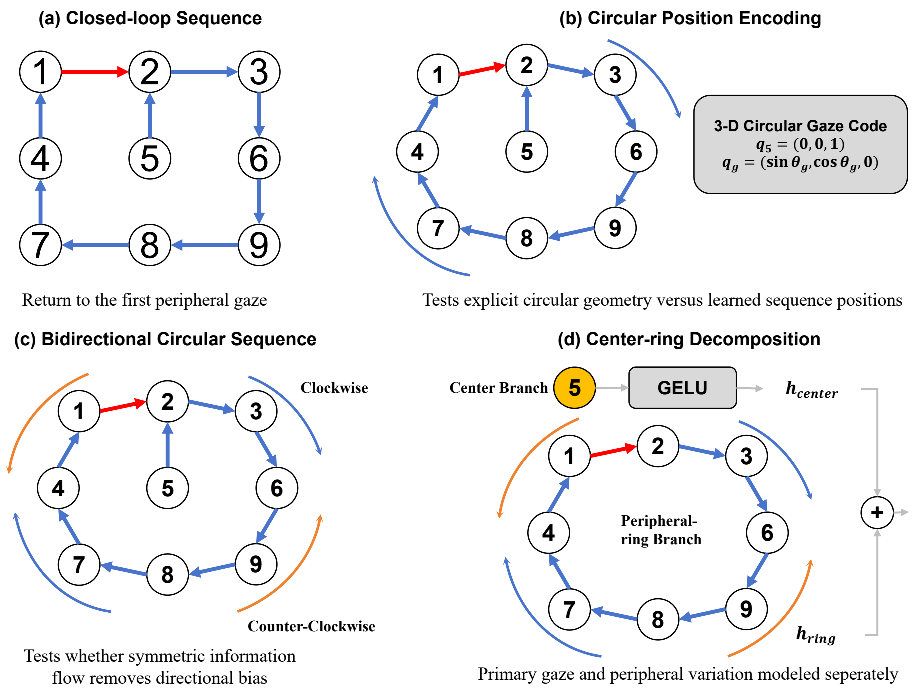

Figure 4. Structural sequence variants evaluated in addition to the linear acquisition trajectory. These variants modify sequence construction or fusion while retaining the same gaze-wise visual features and diagnostic objective.

4.5 Structural Ablation

We next test whether the acquisition trajectory benefits from additional geometric constraints. **Figure 4** defines four variants. Closed-loop repeats the first peripheral gaze at the end of the acquisition sequence. Circular positional encoding represents peripheral locations with projected sine and cosine coordinates. Bidirectional modeling averages representations from clockwise and counter-clockwise circular sequences. Center-ring modeling encodes the primary position separately and fuses it with a bidirectional peripheral ring. The reference row, denoted *linear acquisition*, is the complete GazeMamba sequence model without any of these additional constraints.

Table 4. Structural sequence ablation using the same gaze-wise features and diagnostic objective.

Sequence construction	Accuracy (%)	Weighted F1 (%)
Linear acquisition (Ours)	**59.65**	**54.76**
Closed loop	57.31	52.14
Circular positional encoding	53.80	50.64
Bidirectional circular	54.97	48.94
Center-ring fusion	56.73	53.05

As shown in **Table 4**, the reference model is not a configuration in which no components are used. It already contains shared gaze-wise encoding, ordered state-space propagation, residual Mamba blocks, sequence pooling, and diagnostic heads. This ablation instead evaluates whether additional circular or branch-specific structural priors can improve the complete linear-acquisition model. None of the evaluated variants surpasses the reference model on the present dataset. Closed-loop processing reduces accuracy by 2.34 percentage points, while circular positional encoding produces a reduction of 5.85 points. Center-ring fusion recovers part of this performance but remains 2.92 points below the linear-acquisition model. One possible explanation is that the nine-token sequence is relatively short and that the imposed geometric symmetry does not fully correspond to the unequal discriminative importance of different gaze transitions. Consequently, the simpler acquisition-ordered trajectory provides the most favorable balance between structural bias and model complexity in this study.

4.6 Representation Visualization

To inspect sequence geometry, we extract 192-dimensional held-out representations from the mean-pooling, row-major, fixed-random, and acquisition-ordered checkpoints. Each representation is independently standardized, reduced to 50 dimensions by PCA, and embedded with t-SNE (perplexity 30). Because each panel is fitted separately, only qualitative comparisons should be made across panels.

Compared with mean pooling, acquisition ordering produces more compact clusters for several esotropia and DVD samples, consistent with its higher weighted and macro F1. However, substantial overlap remains among the three exotropia subtypes. Thus, the visualization suggests improved global representation structure rather than complete subtype separation, which is further examined with the confusion matrix.

4.7 Subtype Error Analysis

The normalized confusion matrix in **Figure 6** reveals substantial variation in class-wise recall. Esotropia and exotropia without A/V pattern achieve recalls of 0.77 and 0.82, respectively, whereas DVD, esotropia with V pattern, exotropia

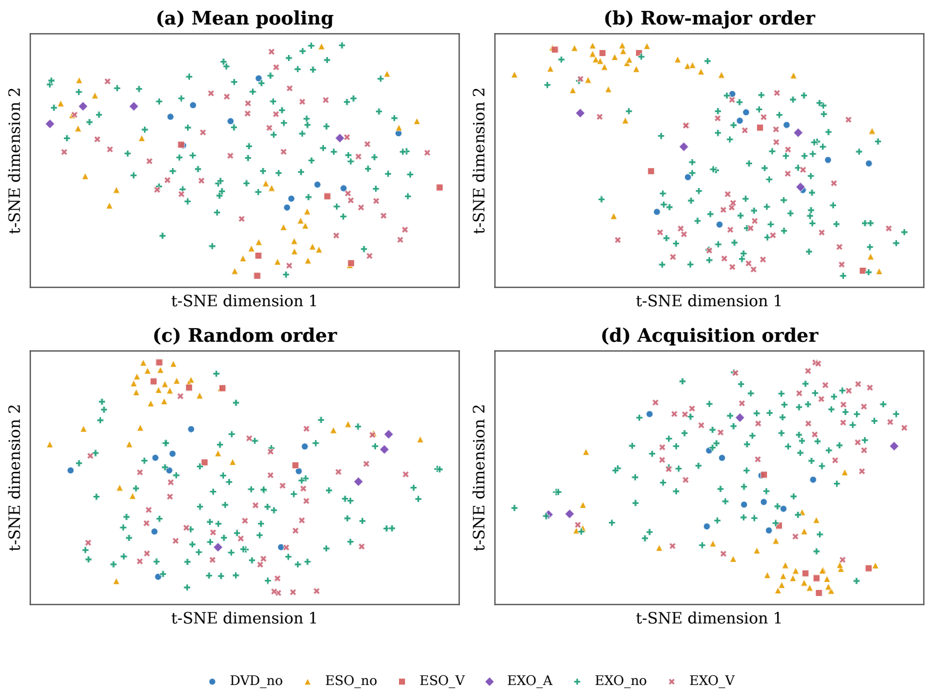

Figure 5. t-SNE visualization of 192-dimensional representations for all 171 held-out samples from (a) mean pooling, (b) row-major order, (c) fixed random order, and (d) acquisition order. Colors and markers denote the six strabismus subtypes. Each panel is embedded independently, so the visualization is qualitative and complements the quantitative metrics.

with A pattern, and exotropia with V pattern reach only 0.10, 0.00, 0.00, and 0.28. Most errors are assigned to exotropia without A/V pattern, explaining the gap between macro and weighted F1.

These results suggest that sequence modeling improves the shared representation but does not fully address subtype imbalance, particularly for rare A/V-pattern and vertical-deviation cases. Class-balanced sampling, cost-sensitive learning, and validation on larger pattern-specific cohorts are therefore important directions for future work.

5 Conclusion

We presented GazeMamba, a gaze-level state-space model for strabismus subtype classification from the photograph of the nine cardinal positions of gaze. The method represents each complete gaze view as a semantic token and separates gaze-wise visual encoding from inter-gaze dependency modeling. With identical gaze-wise features, GazeMamba substantially improves over unordered mean pool-

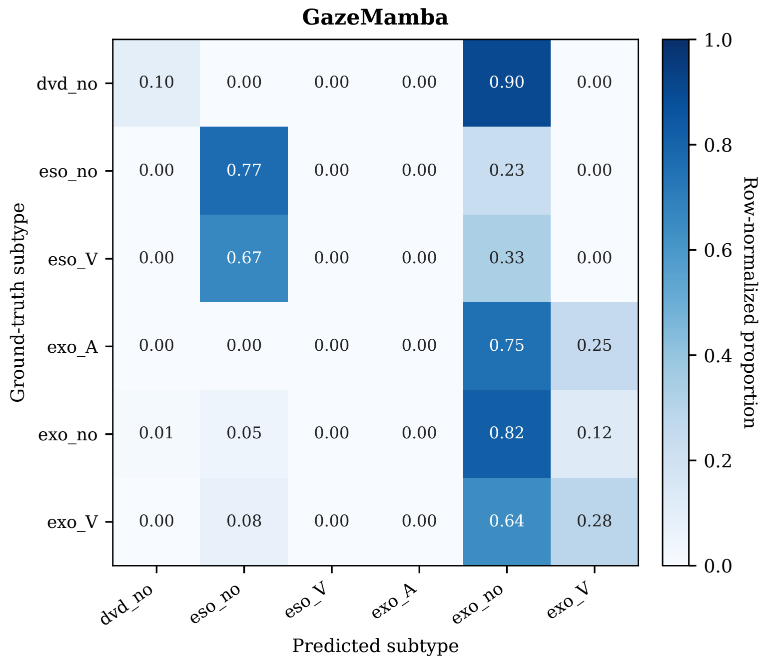

Figure 6. Normalized confusion matrix for GazeMamba on the 171-sample held-out test set. Rows correspond to ground-truth labels and columns to predictions, with each row normalized to sum to one. Values are encoded by the colorbar. The six classes combine deviation type with the without-A-or-V-pattern, A-pattern, and V-pattern attributes. DVD is treated as the vertical-strabismus category.

ing, demonstrating that relations among gaze states contain useful classification evidence. The complete model also improves over the evaluated composed-image backbones. Order and structural analyses show that the acquisition trajectory provides a favorable semantic traversal, whereas additional circular and bidirectional constraints do not improve the present model. These findings motivate further evaluation with same-protocol structured baselines, larger multi-center cohorts, and more balanced representation of pattern-specific subtypes.

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Disclosure of Interests. The authors declare no competing interests relevant to this work.

References

1. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929 (2020)
2. Garg, S., Sunkarapalli, A., Ghosh, D., Kaur, S., Sukhija, J.: Strabnet-cq: an integrated deep learning framework for automated strabismus classification and quantification using ocular landmark detection. BMC Ophthalmology **25**(1), 595 (2025)
3. Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. arXiv preprint arXiv:2312.00752 (2023)
4. Gu, A., Goel, K., Ré, C.: Efficiently modeling long sequences with structured state spaces. arXiv preprint arXiv:2111.00396 (2021)
5. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 770–778 (2016)
6. Huang, G., Liu, Z., Van Der Maaten, L., Weinberger, K.Q.: Densely connected convolutional networks. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2261–2269. IEEE (2017)
7. Huang, X., Lee, S.J., Kim, C.Z., Choi, S.H.: An improved strabismus screening method with combination of meta-learning and image processing under data scarcity. PLoS One **17**(8), e0269365 (2022)
8. Ilse, M., Tomczak, J., Welling, M.: Attention-based deep multiple instance learning. In: International Conference on Machine Learning (ICML). pp. 2127–2136. PMLR (2018)
9. Kang, Y.C., Yang, H.K., Kim, Y.J., Hwang, J.M., Kim, K.G.: Automated mathematical algorithm for quantitative measurement of strabismus based on photographs of nine cardinal gaze positions. BioMed Research International **2022**(1), 9840494 (2022)
10. Karaaslan, Ş., Kobat, S.G., Gedikpınar, M.: A new method based on deep learning and image processing for detection of strabismus with the hirschberg test. Photodiagnosis and Photodynamic Therapy **44**, 103805 (2023)
11. Lee, J., Lee, Y., Kim, J., Kosiorek, A., Choi, S., Teh, Y.W.: Set transformer: A framework for attention-based permutation-invariant neural networks. In: International Conference on Machine Learning (ICML). pp. 3744–3753. PMLR (2019)
12. Lee, S., Choi, J., Kim, H.J.: Efficientvim: Efficient vision mamba with hidden state mixer based state space duality. In: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 14923–14933. IEEE (2025)
13. Liu, L., Zhang, M., Yin, J., Liu, T., Ji, W., Piao, Y., Lu, H.: Defmamba: Deformable visual state space model. In: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 8838–8847. IEEE (2025)

14. Liu, Y., Tian, Y., Zhao, Y., Yu, H., Xie, L., Wang, Y., Ye, Q., Jiao, J., Liu, Y.: Vmamba: Visual state space model. Advances in Neural Information Processing Systems (NeurIPS) **37**, 103031–103063 (2024)
15. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin transformer: Hierarchical vision transformer using shifted windows. In: 2021 IEEE/CVF International Conference on Computer Vision (ICCV). pp. 9992–10002. IEEE (2021)
16. Ma, J., Li, F., Wang, B.: U-mamba: Enhancing long-range dependency for biomedical image segmentation. arXiv preprint arXiv:2401.04722 (2024)
17. Selby, K.A., Rashid, A., Kobyzev, I., Rezagholizadeh, M., Poupart, P.: Learning functions on multiple sets using multi-set transformers. In: Uncertainty in Artificial Intelligence. pp. 1760–1770. PMLR (2022)
18. Tang, X., Wang, Y., Zhuang, J., Luo, L., Guo, J., Xie, X., Liu, J., Wei, P., Zhong, L., Kang, X., et al.: Decis: Dual-evidence corrective verification for interpretable strabismus diagnostic decision-making. arXiv preprint arXiv:2606.09249 (2026)
19. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. Advances in Neural Information Processing Systems (NeurIPS) **30** (2017)
20. Xing, Z., Ye, T., Yang, Y., Liu, G., Zhu, L.: Segmamba: Long-range sequential modeling mamba for 3d medical image segmentation. arxiv 2024. arXiv preprint arXiv:2401.13560 (2025)
21. Yarkheir, M., Sadeghi, M., Azarnoush, H., Akbari, M.R., Khalili Pour, E.: Automated strabismus detection and classification using deep learning analysis of facial images. Scientific Reports **15**(1), 3910 (2025)
22. Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R.R., Smola, A.J.: Deep sets. Advances in Neural Information Processing Systems (NeurIPS) **30** (2017)
23. Zheng, J., Luo, L., Zhuang, J., Wei, P., Zhong, L., Xie, X., Guo, J., Xie, M., Kang, X., Cen, J., et al.: Causality-inspired graph neural network for interpretable strabismus subtype classification. In: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). pp. 107–116. Springer (2025)
24. Zhu, L., Liao, B., Zhang, Q., Wang, X., Liu, W., Wang, X.: Vision mamba: Efficient visual representation learning with bidirectional state space model. arXiv preprint arXiv:2401.09417 (2024)